

PURE

# TRIGONOMETRY

$\frac{2}{\pi}$  /  $\frac{3}{2}$  (reciprocal)

## Reciprocal Trig Ratios

$$\frac{1}{\sin \theta} = \operatorname{cosec} \theta$$

↑ cosecant

$$\frac{1}{\cos \theta} = \sec \theta$$

↑ secant

$$\frac{1}{\tan \theta} = \cot \theta$$

↑ cotangent

Eg Use calculator to find  $\sec 61^\circ$

$$\sec 61^\circ = \frac{1}{\cos 61^\circ} = 2.06 \text{ (3sf)}$$

$$\cot 45^\circ = \frac{1}{\tan 45^\circ} = 1$$

• solve for  $0 \leq x \leq 90^\circ$

$$\operatorname{cosec} x = 5$$

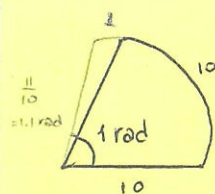
$$\frac{1}{\sin x} = 5$$

$$\sin x = \frac{1}{5}$$

$$x = \sin^{-1}\left(\frac{1}{5}\right)$$

$$x = 11.5 \text{ (1.d.p.)}$$

1.d.p



1 radian = arc length = r

$$s = r\theta$$

$$\theta = \frac{s}{r} = \frac{11}{10}$$

$$= \frac{2\pi r}{r} = 2\pi = 360^\circ$$

$$\text{Arc length} = \frac{\pi r \theta}{180} = r\theta$$

$$A_{\text{sector}} = \frac{\pi r^2 \theta}{360} = \frac{1}{2} r^2 \theta$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 - \cos^2 \theta = (1 - \cos \theta)(1 + \cos \theta)$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

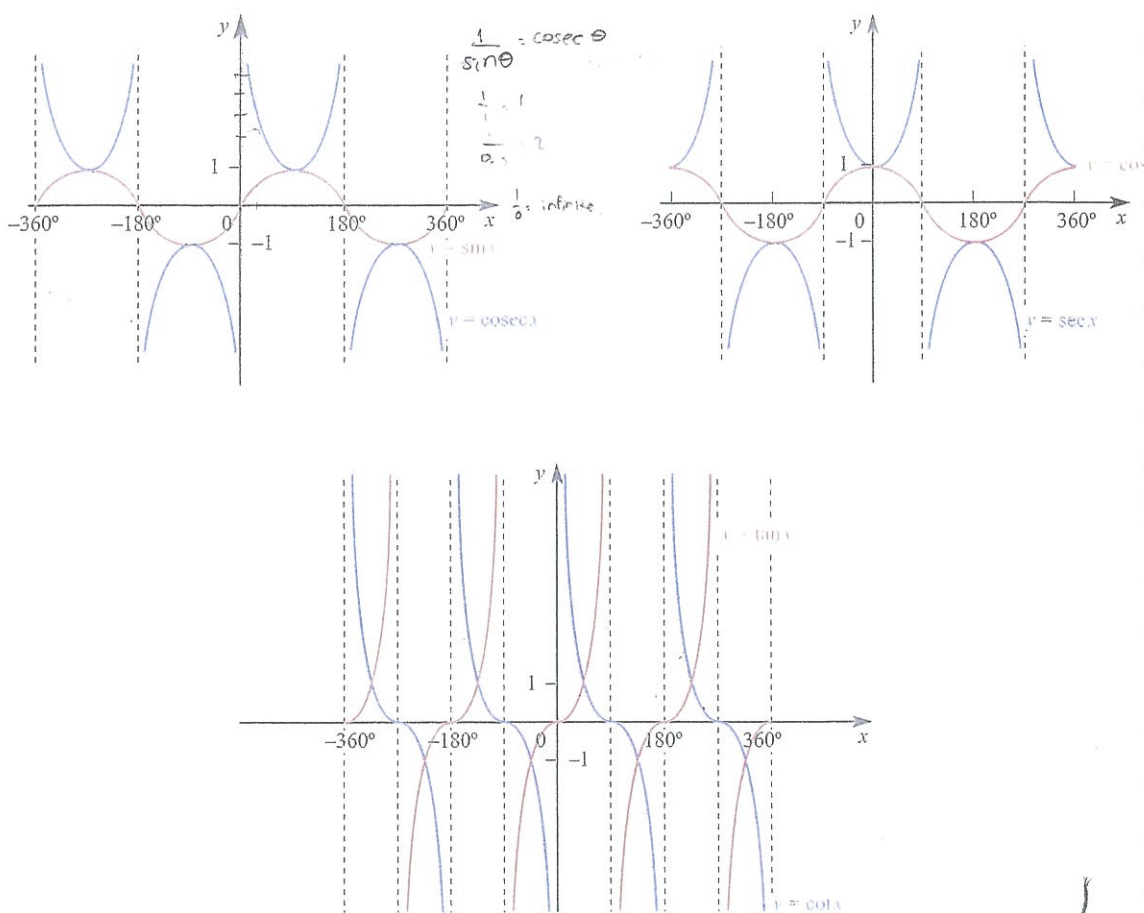
## Reciprocal trigonometrical functions

As well as the three main trigonometrical functions, there are three more which are commonly used. These are their reciprocals – cosecant (cosec), secant (sec) and cotangent (cot), defined by

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}; \quad \sec \theta = \frac{1}{\cos \theta}; \quad \cot \theta = \frac{1}{\tan \theta} \left( = \frac{\cos \theta}{\sin \theta} \right).$$

Each of these is undefined for certain values of  $\theta$ . For example,  $\operatorname{cosec} \theta$  is undefined for  $\theta = 0^\circ, 180^\circ, 360^\circ, \dots$  since  $\sin \theta$  is zero for these values of  $\theta$ .

Figure 3.2 shows the graphs of these functions. Notice how all three of the functions have asymptotes at intervals of  $180^\circ$ . Each of the graphs shows one of the main trigonometrical functions as a red-line and the related reciprocal function as a blue line.





# Trig Identities with Sec, cosec, Cot

We know

$$\frac{\sin^2 x}{\sin^2 x} + \frac{\cos^2 x}{\sin^2 x} = \frac{1}{\sin^2 x}$$

$$\downarrow \qquad \downarrow$$
$$1 + \frac{1}{\tan^2 x} = \text{cosec}^2 x$$

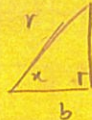
$$1 + \cot^2 x = \text{cosec}^2 x$$

We know

$$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\sin^2 x + \cos^2 x = 1$$



$$\sin x = \frac{a}{r}$$

$$a = \sin x \cdot r$$

$$\cos x = \frac{b}{r}$$

$$r^2 = (\sin x \cdot r)^2 + (\cos x \cdot r)^2 = \sin^2 x \cdot r^2 + \cos^2 x \cdot r^2$$

$$r^2 = \sin^2 x \cdot r^2 + \cos^2 x \cdot r^2$$

$$1 = \sin^2 x + \cos^2 x$$

$$\tan x = \frac{\sin x \cdot r}{\cos x \cdot r} = \frac{\sin x}{\cos x}$$

Example application:

(a) Express  $3 \tan^2 \theta - \sec \theta$  in terms of  $\sec \theta$  only.

$$\begin{aligned} \text{We have } \tan^2 \theta &= \sec^2 \theta - 1 \\ 3(\sec^2 \theta - 1) - \sec \theta &= 3 \sec^2 \theta - 3 - \sec \theta \\ &= 3 \sec^2 \theta - \sec \theta - 3 \end{aligned}$$

(b) Hence solve  $3 \tan^2 \theta - \sec \theta = 1$  for  $0 \leq \theta < 360^\circ$

$$3 \sec^2 \theta - \sec \theta - 3 = 1$$

$$3 \sec^2 \theta - \sec \theta - 4 = 0$$

$$(3 \sec \theta - 4)(\sec \theta + 1) = 0$$

$$(3 \sec \theta - 4)(\sec \theta + 1) = 0$$

$$\text{So, } \sec \theta = \frac{4}{3} \quad \text{or} \quad \sec \theta = -1$$

$$\bullet \sec \theta = \frac{4}{3}$$

or

$$\sec \theta = -1$$

$$\frac{1}{\cos \theta} = \frac{4}{3}$$

$$\frac{1}{\cos \theta} = -1$$

$$\cos \theta = -1$$

$$\cos \theta = \frac{3}{4}$$

$$\theta = \cos^{-1}\left(\frac{3}{4}\right)$$

$$\theta = 180^\circ$$

$$\frac{S}{A} = 41.4096 \dots$$

$$\frac{T}{C} = 41.4$$

or

$$\theta = 360^\circ - 41.409 \dots$$

$$= 318.5903 \dots$$

$$= 318.6$$



$$\text{So, } \theta \in \{41.4^\circ, 180^\circ, 318.6^\circ\}$$



# The Compound Angle Formulae

If A and B are angles

$$\sin(A + B) \equiv \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) \equiv \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) \equiv \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) \equiv \cos A \cos B + \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

use formula to find exact values.

Eg  $\sin 75^\circ = \sin(45^\circ + 30^\circ)$

30, 45, 60, 90

$$\begin{aligned} &= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ \\ &= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \times \frac{1}{2} \\ &= \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4} \end{aligned}$$

Eg.  $\cos 15^\circ = \cos(60^\circ - 45^\circ)$

$$\begin{aligned} &= \cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ \\ &= \frac{1}{2} \times \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} \\ &= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4} \end{aligned}$$

or  $\cos(45^\circ - 30^\circ)$

$$\begin{aligned} &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \times \frac{1}{2} \\ &= \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6} + \sqrt{2}}{4} \end{aligned}$$

$\tan 75^\circ = \tan(30^\circ + 45^\circ)$

$$\begin{aligned} &= \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ} = \frac{\frac{\sqrt{3}}{3} + 1}{1 - \frac{\sqrt{3}}{3}} = \frac{\frac{\sqrt{3} + 3}{3}}{\frac{3 - \sqrt{3}}{3}} \end{aligned}$$

$$= \frac{\sqrt{3} + 3}{3 - \sqrt{3}} \cdot \frac{(\sqrt{3} + 3)(3 + \sqrt{3})}{3^2 - \sqrt{3}^2} = \frac{\sqrt{3}^2 + 2 \times 3 \times \sqrt{3} + 3^2}{6} = \frac{3 + 6\sqrt{3} + 9}{6} = \frac{12 + 6\sqrt{3}}{6}$$

### Further Example :

If  $\sin A = \frac{8}{17}$  and  $\sin B = \frac{12}{13}$  ( $0-90^\circ$   $90-180^\circ$  B is acute, A is obtuse)

Find the exact value of  $\tan(A+B)$

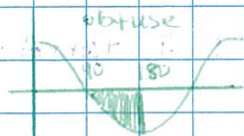
⇒ Answer :  $\tan(A+B) = \frac{\sin(A+B)}{\cos(A+B)}$



$$\frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B}$$

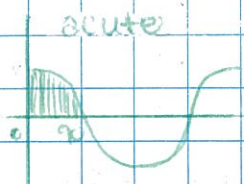
$$\cos A = \pm \sqrt{1 - \sin^2 A}$$

$$= \pm \sqrt{1 - \left(\frac{8}{17}\right)^2} = \pm \frac{15}{17}$$
$$= -\frac{15}{17}$$



$$\cos B = \pm \sqrt{1 - \sin^2 B}$$

$$= \pm \sqrt{1 - \left(\frac{12}{13}\right)^2} = \pm \frac{5}{13}$$
$$= \frac{5}{13}$$



$$\text{so, } \tan(A+B) = \frac{\left(\frac{8}{17}\right)\left(\frac{5}{13}\right) + \left(-\frac{15}{17}\right)\left(\frac{12}{13}\right)}{\left(-\frac{15}{17}\right)\left(\frac{5}{13}\right) - \left(\frac{8}{17}\right)\left(\frac{12}{13}\right)} = \frac{140}{171}$$



(i) show that  $\tan(45^\circ + x) - \tan x = 2$

can be written as

$$\tan^2 x + 2 \tan x - 1 = 0$$

(ii). Hence, solve

$$\tan(45^\circ + x) - \tan x = 2 \quad \text{for } 0 \leq x \leq 180^\circ$$

$\Rightarrow$  Answer

$$(i) \quad \tan(45^\circ + x) - \tan x = 2$$

$$\frac{\tan 45^\circ + \tan x}{1 - \tan 45^\circ \tan x} - \tan x = 2$$

$$\frac{1 + \tan x}{1 - \tan x} - \tan x = 2$$

$$1 + \tan x - \tan x(1 - \tan x) = 2(1 - \tan x)$$

$$1 + \tan x - \tan x + \tan^2 x = 2 - 2 \tan x$$

$$\tan^2 x + 2 \tan x - 1 = 0$$

$$(ii) \quad \text{Let } \tan x = A \quad A^2 + 2A - 1 = 0$$

$$A = \frac{-2 \pm \sqrt{2^2 - 4(1)(-1)}}{2}$$

$$A = -1 + \sqrt{2} \\ = 0.414 \dots$$

$$\text{or } A = -1 - \sqrt{2}$$

$$= -2.414 \dots$$

|   |   |
|---|---|
| S | A |
| T | C |

$$\tan x = 0.414$$

$$x = \tan^{-1}(0.414)$$

$$= 22.4895 \dots$$

$$= 22.5$$

$$\tan x = -2.414$$

$$x = \tan^{-1}(2.414) = 67.4982 \dots$$

$$x = 180^\circ - 67.4982 \dots$$

$$= 112.5$$

$$x \in \{22.5 \text{ or } 112.5\}$$

# Double Angle Formula

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 2 \cos^2 A - 1 \\ &= 1 - 2 \sin^2 A\end{aligned}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin(A+A) = \sin A \cos A + \cos A \sin A$$

$$\begin{aligned}\cos(A+A) &= \cos A \cos A - \sin A \sin A \\ \cos^2 A - \sin^2 A &= \cos^2 A - (1 - \cos^2 A)\end{aligned}$$

Eg. Solve  $2 \sin 2A = \cos A$

$$0 \leq A < 360^\circ$$

$$2(2 \sin A \cos A) = \cos A$$

$$4 \sin A \cos A = \cos A$$

$$4 \sin A \cos A - \cos A = 0$$

$$\cos A (4 \sin A - 1) = 0$$

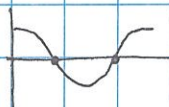
So,  $\cos A = 0$

or

$$\sin A = \frac{1}{4}$$

$$A = 90^\circ \text{ or } 270^\circ$$

you cannot do  $\frac{4 \sin A \cos A}{\cos A} = \frac{\cos A}{\cos A}$   
because it gives only  $\frac{1}{2}$  the  
( $x^2 = x$ )



$$A = \sin^{-1}\left(\frac{1}{4}\right)$$



$$A = 14.4775 \dots$$

$$= 14.5$$

or

$$A = 180^\circ - 14.477 \dots$$

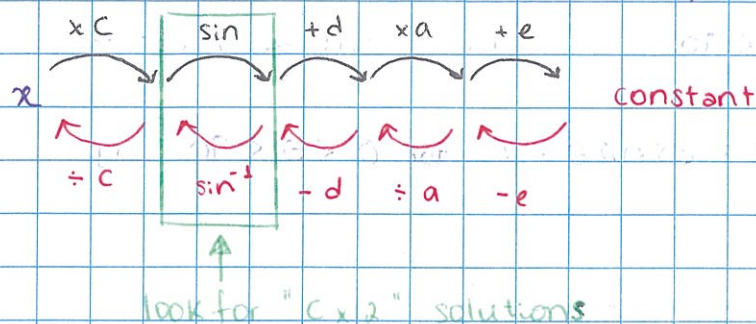
$$= 165.5224 \dots$$

$$= 165.5^\circ$$

$$A \in \{14.5^\circ, 90^\circ, 165.5^\circ, 270^\circ\}$$



## Solving $a(\sin cx + d) + e = \text{constant}$



## Adding / Subtracting sin / cos waves

These formulas are not on the formula sheet

max amplitude when added  $a \sin x$  and  $b \cos x$

$$a \sin x + b \cos x = R \sin(x + \alpha) \quad \text{where } a > 0$$

$\rightarrow$  how much it shifts

$b > 0$  constants

$$a \sin x - b \cos x = R \sin(x - \alpha) \quad R > 0$$

$$a \cos x + b \sin x = R \cos(x - \alpha) \quad 0 < \alpha < 90^\circ$$

$$0 < \alpha < \frac{\pi}{2}$$

$$a \cos x - b \sin x = R \cos(x + \alpha) \quad \text{and } R = \sqrt{a^2 + b^2}$$

$$\alpha = \tan^{-1}\left(\frac{b}{a}\right)$$

eg.  $2 \sin x + 3 \cos x = R \sin(x + \alpha)$

where  $R = \sqrt{2^2 + 3^2}$

$$R = \sqrt{13}$$

$$R = 3.61 \text{ (3 s.f.)}$$

$$\alpha = \tan^{-1}\left(\frac{3}{2}\right)$$

$$\alpha = 56.3^\circ$$

$$\tan \alpha = \frac{b}{a} \quad \left\{ \begin{array}{l} \sin \alpha = \frac{b}{R} \\ \cos \alpha = \frac{a}{R} \end{array} \right.$$

Cambridge Q:

(i) Express  $\cos x + 3\sin x$  as  $R \cos(x - \alpha)$  [3]  
where  $R > 0$   $0^\circ < \alpha < 90^\circ$

(ii) Hence to solve  $\cos 2\theta + 3\sin 2\theta = 2$  for  $0^\circ < \theta < 90^\circ$  [5]

Ans:

(i).  $R = \sqrt{1^2 + 3^2} = \sqrt{10}$

$$\alpha = \tan^{-1}\left(\frac{3}{1}\right)$$

$$\alpha = 71.565\dots$$

$$\alpha = 71.6^\circ \text{ (3sf)}$$

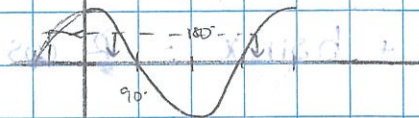
(ii).  $\cos x + 3\sin x = \sqrt{10} \cos(x - 71.6^\circ)$

$$\cos 2\theta + 3\sin 2\theta = \sqrt{10} \cos(2\theta - 71.6^\circ)$$

Solve  $\sqrt{10} \cos(2\theta - 71.6^\circ) = 2$

$$\cos(2\theta - 71.6^\circ) = \frac{2}{\sqrt{10}}$$

$$2\theta - 71.6^\circ = \cos^{-1}\left(\frac{2}{\sqrt{10}}\right)$$



$$2\theta - 71.6^\circ = 50.8^\circ \quad \text{or} \quad 2\theta - 71.6^\circ = -50.8^\circ$$

$$\theta = \frac{50.8 + 71.6}{2}$$

$$\theta = \frac{-50.8 + 71.6}{2}$$

$$\theta = 61.2$$

$$\theta = 10.4$$



# TRIG PROOFS

Example: Prove that  $\frac{\sec x}{\tan x} = \operatorname{cosec} x$

my answer:

$$\text{LHS} = \frac{\sec x}{\tan x}$$

$$= \frac{\sec x}{\tan x}$$

$$= \frac{1}{\cos x} \div \frac{\sin x}{\cos x}$$

$$= \frac{1}{\cos x} \times \frac{\cos x}{\sin x}$$

$$= \frac{1}{\sin x}$$

$$= \operatorname{cosec} x$$

$$= \text{RHS}$$

Some techniques:

- Start with the more complicated side
- write (where possible) in terms of  $\sin$  and/or  $\cos$
- use simple algebra
  - expanding
  - factorising
  - cancelling
- where + or - fractions  
find a common denominator
- use all possible identities from the formula sheet  
eg  $\cos^2 \theta = 1 - \sin^2 \theta$ ,  
 $\sin(2\theta) = 2 \sin \theta \cos \theta$ , etc.

## PRINCIPAL VALUES OF $\sin^{-1}$ , $\cos^{-1}$ , $\tan^{-1}$

The principal value is the value that your calculator gives for  $\sin^{-1}$ ,  $\cos^{-1}$ ,  $\tan^{-1}$ . This can then be used to determine other values for  $\sin^{-1}$  etc.

$$-\frac{1}{2}\pi \leq \sin^{-1} x \leq \frac{1}{2}\pi$$

$$0 \leq \cos^{-1} x \leq \pi$$

$$-\frac{1}{2}\pi < \tan^{-1} x < \frac{1}{2}\pi$$